

MOVIT

Move Smarter. Wait Less.

PROJECT DOCUMENTATION

A Smart Public Transportation Web Application

Project	MOVIT
Domain	Smart Transportation / Web Application
Project Type	Functional Web Prototype
Institution	Mohan Babu University
Department	Computer Science and Engineering
Academic Year	2026–27

“The right bus. Not just the next bus.”

1. Introduction

MOVIT is a smart public-transport web application that helps passengers choose a suitable bus for their journey instead of simply taking the next available bus. The project focuses on bringing useful travel information into one place so that passengers can compare their options before making a decision.

2. Problem Statement

Passengers often make bus decisions with limited information. They may know when one bus is arriving, but may not know whether another bus is arriving soon, whether the first bus is delayed, or how crowded it is. Because of this, taking the first bus is not always the best choice.

MOVIT tries to solve this problem by showing more information about the available buses and helping the passenger choose an option that better fits the journey.

3. Objectives

- Show available buses arriving within the next 60 minutes.
- Display ETA and possible delays.
- Show crowding or estimated occupancy information.
- Provide transport status and route updates.
- Allow passengers to submit reports about their journey.
- Compare available buses using important travel conditions.
- Highlight a suitable or optimal bus option.
- Reduce unnecessary waiting and help passengers make better travel decisions.

4. Proposed System

The proposed system gives passengers a simple way to search for buses between two locations and compare the available options. Instead of focusing only on the earliest arriving bus, MOVIT considers several conditions that can affect the passenger's choice.

The main flow is:

Journey → Find a Bus → Compare Options → Check Transport Status → Travel

5. Main Features

5.1 Journey Planning

Users select their FROM bus stop and TO bus stop. The selected journey is then used to find suitable bus options. The prototype demonstrates an Andhra Pradesh journey example such as Tirupati → Chittoor.

5.2 Find a Bus

MOVIT shows bus arrivals for the next 60 minutes. This allows users to see more than one possible option instead of automatically taking the first bus.

5.3 Smart Bus Decision

The main idea is to choose the bus that is more suitable for the journey, not simply the bus that arrives first. The recommendation can consider arrival time, waiting time, delays, crowding and route status.

5.4 Transport Status

Passengers can check information such as delays, crowding conditions, route changes and other transport-related updates before travelling.

5.5 Passenger Report

Passengers can share information about the conditions they experience. Reports can include crowding, delay, breakdowns, route changes and comments. This information can be useful to other passengers.

5.6 Real-Time Data Concept

The project can be extended to use live data from GPS/location information, electronic ticketing or boarding data, occupancy sensors and passenger reports. These sources could help provide more accurate information in a real-world deployment.

6. System Workflow

1. Open MOVIT
2. Select the FROM location
3. Select the TO location
4. Search for available buses
5. View buses arriving within the next 60 minutes
6. Compare arrival time, delay, crowding and route status
7. View the suggested bus
8. Check transport status if required
9. Submit a passenger report when appropriate

7. User Experience

The website is designed around a simple journey flow so that passengers can get useful information without going through a complicated process. The main sections are journey selection, bus options, transport status and passenger reporting.

8. Current Prototype

The current version is a functional web prototype using demo or simulated transport data. It demonstrates the main idea and user flow but should not be treated as a fully deployed live transport-authority system.

9. Testing

Test	Expected Result	Status
Select FROM and TO	Bus options are displayed	Passed
View next 60 minutes	Upcoming buses are shown	Passed

Check bus information	ETA, delay and crowding are visible	Passed
Open transport status	Status information is available	Passed
Submit passenger report	Report option is available	Passed

10. Limitations

- The current prototype uses demo or simulated transport information.
- Live GPS tracking is not demonstrated as a fully deployed feature.
- Passenger reports may need validation in a production system.
- The quality of the recommendation depends on the available transport data.

11. Future Scope

- Real-time GPS and live bus tracking.
- Real-time crowd estimation.
- Push notifications for selected buses.
- Public transport API integration.
- Personalized route preferences.
- Historical travel information.
- Improved bus recommendation methods.
- Mobile application.
- User accounts and profiles.
- Better validation of passenger reports.

12. Conclusion

MOVIT focuses on a simple problem: passengers do not always need the next bus; they need enough information to choose the better bus for their journey. By bringing arrival time, delay, crowding, route status and passenger updates into one place, MOVIT aims to make everyday bus decisions easier and more informed.

The current project is a functional prototype and provides a base that can later be connected to real-time transport data.

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13. Project Information

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Tagline	Move Smarter. Wait Less.
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Note: Specific programming languages, frameworks, databases, APIs and deployment technologies are not listed here because they were not confirmed in the provided project material.